HW 4

Due on 05 March 2026 at 8PM on Gradescope.

This assignment is worth 25 points, and you should be spending roughly 10 minutes per point on average. If you find yourself spending more than $10n$ minutes on a n-point problem, reach out to me via drop-in hours, Discord, or email.

Collapse all exercises

Goals

- Practice with new definitions
- Equivalence relations
- Normal subgroups
- Quotient groups

▼ EXERCISE 1 [2]

Either save this exercise for last, or do it incrementally as you work on the homework. You may need to add `\usepackage{xcolor}` to the preamble of your document.

1. Once you have completed the assignment, fill out the following template listing collaborators, tools, and a reflection on each problem you worked on.

```
Collaborators: Alex, Ross, Margaret, Joan

\noindent
Tools and resources used: On Exercise 1, I used ChatGPT to help
\begin{enumerate}
  \item[Exercise 1]
    {\color{red} Points attempted: 7}.
    {\color{blue} Time spent: 56 minutes}.
    {\color{brown} Difficulty: 3/10}.
    {\color{green!50!black} Self-assessed score: 6/7}.
    {\color{magenta} Questions/Feedback: None.}
  \item[Choice Exercise 7]
    {\color{red} Points attempted: 3}.
    {\color{blue} Time spent: 18 minutes}.
    {\color{brown} Difficulty: 7/10}.
    {\color{green!50!black} Self-assessed score: 3/3}.
    {\color{magenta} Questions/Feedback: How should I be thinki
\end{enumerate}
```

2. Choose one or two problems that you would most like feedback on, with a brief note of what you would most like me to focus on. (This could be "mathematical correctness," "grammar," "notation," "formatting," or anything else.)

▼ EXERCISE 2 [7]

There are some ideas from class that in hindsight I wish I had emphasized more. In this exercise, you will self-teach ideas that we glanced over in class.

1. We *defined* the dihedral group in terms of its group presentation:

$$D_n = \langle r, f \mid r^n = f^2 = (rf)^2 = e \rangle.$$

a. Prove that $r^k f = f r^{n-k}$ for all $0 < k < n$.

b. Prove that every element can be written in the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$

c. Prove that the order (number of elements) of the group is $D_n = 2n$.

2. We say that a group $(G, *)$ is an *abelian group* if and only if $a * b = b * a$ for all $a, b \in G$.

a. Prove that if G is *generated* by g_1 and g_2 then G is abelian if and only if $g_1 * g_2 = g_2 * g_1$.

b. Suppose G is *generated* by $g_1, g_2, \ldots, g_n$. Do you think that G is abelian if and only if $g_i * g_j = g_j * g_i$ for all $i, j \in \{1, 2, \ldots, n\}$? Explain your reasoning, but you do not have to provide a proof.

c. Prove that cyclic groups C_n are abelian.

d. Prove that the *general linear group* of 2×2 real-valued matrices

$$GL_2(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} \mid \det(A) \neq 0\}$$

is *not abelian*.

e. Prove that the dihedral group D_n is abelian if and only if $n \leq 2$.

3. The *direct product of groups* is a way of combining two groups into a new group. The notation can look a bit intimidating, but the basic idea is simple: just treat the groups independently. Here's the formal definition, but keep in mind that $*$, $\star$, and $\odot$ are just names for binary operations like $+$, $\times$, and $\circ$.

Given two groups $(G_1, *)$ and $(G_2, \star)$, we define the *direct product of groups* as

$$G_1 \times G_2 = \{(g_1, g_2) \mid g_1 \in G_1 \text{ and } g_2 \in G_2\}$$

with the binary operation, $\odot$, where

$$(g_1, g_2) \odot (g_1', g_2') = (g_1 * g_1', g_2 \star g_2').$$

a. Let $(G_1, *) = (D_4, *)$ be the dihedral group of the square, and let $(G_2, \star) = (S_3, \circ)$ be the symmetric group under function composition. Show that in $G_1 \times G_2$,

$$(f, (132)) \odot (r, (23)) = \left(r^3 f, (13)\right).$$

b. Prove that for any two groups G_1 and G_2, the direct product of groups is a group.
c. Prove that if $G_1 \times G_2$ is abelian if and only if G_1 and G_2 are both abelian.
d. Let $\mathbb{Z}/k\mathbb{Z}$ denote the group of the integers mod k under addition. Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Compute the order of G. Prove that G has an element of order $|G|$.
e. Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Compute the order of G. Prove the G has no element of order $|G|$.

▼ EXERCISE 3 [1]

Post on Discord with your favorite example, resource, homework problem, theorem, or definition that you've seen in MAT 4170 this term.

Choice Exercises

Do 15 points of the following.

▼ CHOICE EXERCISE 4 [5+≤7] (REVIEW/PRACTICE)

1. [5] We sometimes describe groups by their *presentation*, $\langle S \mid R \rangle$, where S is a set of *generators*, and R is a set of *relations*. Recall, we define the dihedral group of the n-gon as

$$D_n = \langle \underbrace{r, f}_{S} \mid \underbrace{r^n = f^2 = (rf)^2 = e}_{R} \rangle.$$

What this means is that every element of $s \in D_n$ can be written as a *word*

$$s = g_1^{e_1} g_2^{e_2} \dots g_m^{e_m},$$

where $g_i \in S = \{r, f\}$, $e_i \in \mathbb{Z}$, *and* where two words are equal if we can modify one to the other using the relations in R.

a. Draw a Cayley diagram of the group with presentation

$$G = \langle r_1, r_2 \mid r_1^2 = r_2^2 = (r_1 r_2)^3 = e \rangle.$$

(Hint: $|G| \leq 10$.)

b. Use the generators and relations to prove that $r^{-3} f r f^3 = r \in D_5$.

c. Prove that for all positive integers k, $(r^k f)^2 = e \in D_n$.

d. Write a group presentation of the cyclic group C_n.

2. [≤7] Recall that an *equivalence* relation $\sim$ on a set X is a relation that is

- Reflexive: $x \sim x$ for all $x \in X$.
- Symmetric: If $x \sim y$ then $y \sim x$.
- Transitive: If $x \sim y$ and $y \sim z$, then $x \sim z$.

Do any subset of the following.

a. [2] State which of the following are equivalence relations and briefly describe why.

i. Let $X = \mathbb{Z}$ be the set of integers, and the relation be *equals* $(=)$.

ii. Let $X = \mathbb{R}$ be the set of real numbers, and the relation be *less than or equal to* $(\leq)$.

iii. Let $X = \mathbb{Q}$ be the set of rational numbers, and the relation be *less than* $(<)$.

iv. Let $X = D_4$ be the set of symmetries of the square, and the relation be *not equal to* $(\neq)$.

b. [1] Suppose we have a function $f\colon X \to Y$ the relation R is defined as

$$x_1 \; R \; x_2 \iff f(x_1) = f(x_2).$$

Prove or disprove that R is an equivalence relation.

c. [1] Consider the subgroup $H \leq (G, *)$ and define the relation R as

$$a \; R \; b \iff a^{-1} * b \in H.$$

Prove or disprove that R is an equivalence relation.

d. [1] Consider the subgroup $H \leq (G, *)$ and define the relation R as

$$a \; R \; b \iff a * b \in H.$$

Prove or disprove that R is an equivalence relation.

e. [2] Let $(G, *)$ be a group and define the relation R as

$$a \; R \; b \iff a = g * a * g^{-1} \text{ for some } g \in G.$$

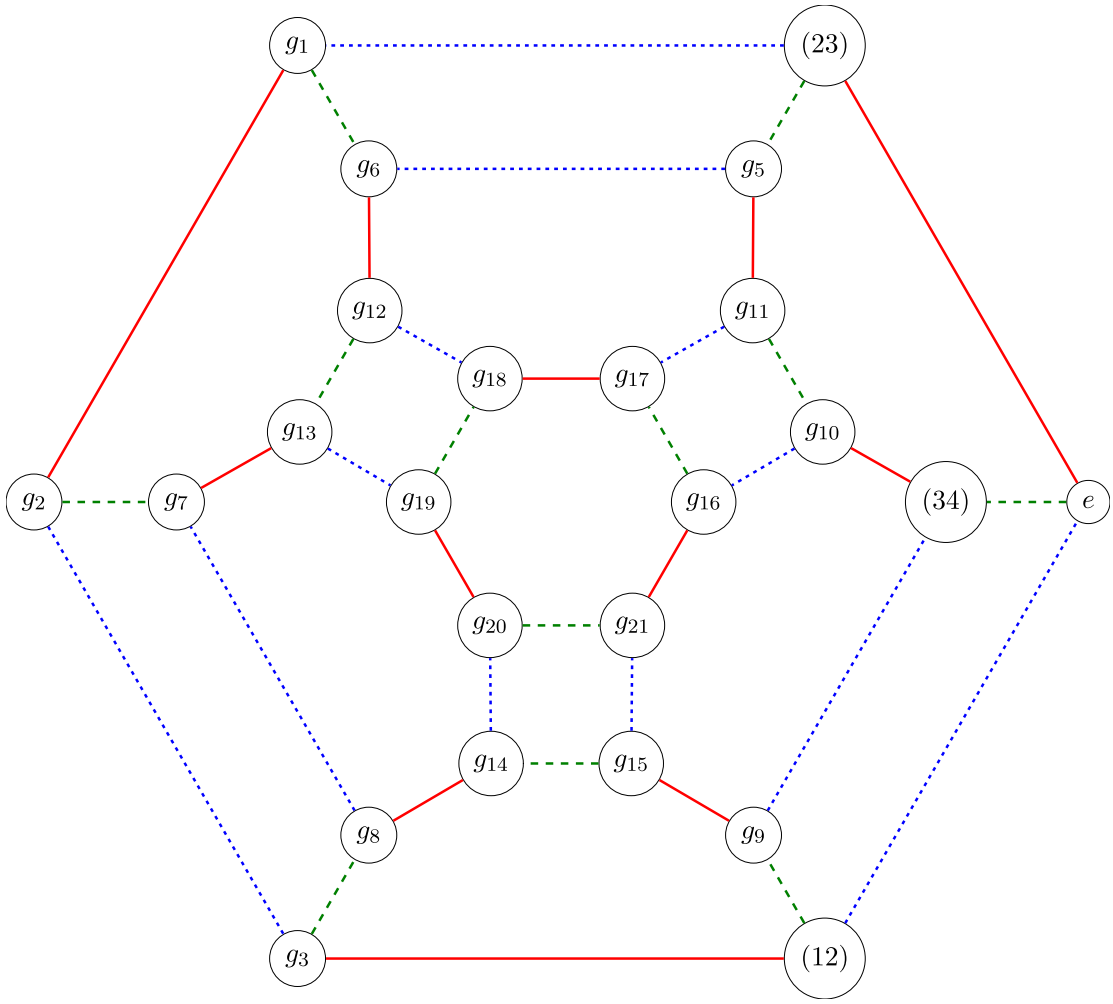
Prove or disprove that R is an equivalence relation.

▼ CHOICE EXERCISE 5 [4 + ≤4 + 4] 🌟🌟🌟

The following three parts were all noted as students' favorite problems in Fall 2024.

1. In this part, you will draw my favorite Cayley diagram.

a. In the Cayley diagram for S_4 below, four of the vertices have been specified: e, (12), (23), and (34). Use this to determine what the three different edges ("arrows") correspond to.



Cayley diagram for S_4 generated by "adjacent transpositions"

b. This Cayley graph (i.e. Cayley diagram) is called a "polytope" because the vertex/edge configuration can be mapped onto the vertices and edges of a polyhedron. Go look up the list of the 13 Archimedean solids, and determine which solid it corresponds to and state why.[1]

c. Edit the TikZ code below to change g_i to the resulting permutation in cycle notation for $i \in \{1, 2, \dots, 21\}$.

```
\begin{tikzpicture}[scale=2]
  \node[draw, circle] (G01) at ({3.7*cos(0)},   {3.7*sin(0)})
  \node[draw, circle] (G02) at ({3.7*cos(60)},  {3.7*sin(60)}
  \node[draw, circle] (G03) at ({3.7*cos(120)}, {3.7*sin(120)
  \node[draw, circle] (G04) at ({3.7*cos(180)}, {3.7*sin(180)
  \node[draw, circle] (G05) at ({3.7*cos(240)}, {3.7*sin(240)
```

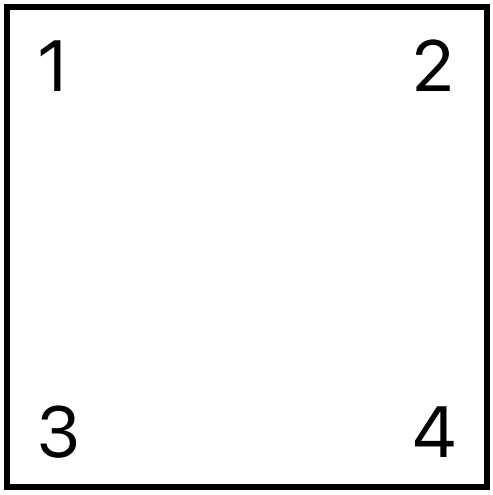
2. We begin with the definition of a *relation*. Given a set X we say a relation R is a subset of the Cartesian product $R \subseteq X \times X$ and we say that $a \; R \; b$ if if $(a, b) \in R$.

For example, if $X = \{1, 2, 3\}$ then if R is the "less than" relation, then formally $R = \{(1, 2), (1, 3), (2, 3)\}$.

Do any subset of the following. You'll get 1 point if you find at least four, 2 points if you find at least six, 3 points if you find exactly seven, and 4 points if you find all eight.

a. Reflexive, symmetric, and transitive.
b. Reflexive, symmetric, and not transitive.
c. Reflexive, not symmetric, and transitive.
d. Reflexive, not symmetric, and not transitive.
e. Not reflexive, symmetric, and transitive.
f. Not reflexive, symmetric, and not transitive.
g. Not reflexive, not symmetric, and transitive.
h. Not reflexive, not symmetric, and not transitive.

3. We can *embed* the dihedral group D_n into S_n. (In fact, Cayley's theorem states that we can embed any finite group G into S_N for large enough some N.) For example, if we mark the corners of the square 1, 2, 3, and 4, then rotating/reflecting the square permutes these numbers.



a. Write down the eight permutations in S_4 (in cycle notation) corresponding to the eight symmetries of the square.
b. We can also write this in terms of the generators of a subgroup. Find two elements $x, y \in S_4$ such that the eight permutations given above are equal to $\langle x, y \rangle$.

▼ CHOICE EXERCISE 6 [≤10] (PROGRAMMING)

1. [3] Write a function that takes a word in S_n and outputs it in cycle notation. For instance, if the word is $\pi = 156243 \in S_n$ then the input to your program would be `[1,5,6,2,4,3]`, and since $\pi = (254)(36)$ in cycle notation, the output would be `[[254],[36]]` or something equivalent.
2. [2] Write a function that goes the other way: it takes in input of a permutation in cycle notation (e.g. `[[254],[36]]`) and outputs the corresponding permutation as a word (e.g. `[1,5,6,2,4,3]`.)
3. [2] Write a function that takes two permutations as words and composes them, working right-to-left as usual.
4. [2] Write a function that takes two permutations in cycle notation and composes them, working right-to-left as usual.
5. [1] Write two functions that compute the order of a permutation, one that works on words, and one that works on cycle notation.

▼ Choice Exercise 7 [≤3 + ≤3] (video/self-study)

1. [≤3] (Let n be the number of points you're claiming for this problem. Clearly state n.)

a. Spend $6n$ minutes exploring Clear Crystal Conundrums: A Multifaceted Intro to Group Theory by Explanaria. This is an interactive explanation illustrating one place where group theory shows up in chemistry and geology.

b. Write n or more sentences about what you understood, didn't understand, liked, were curious about, or any other thoughts you had.

2. [≤3] Choose $n \in \{1, 2, 3\}$, which is the number of points of you will get for this part. Start a timer for $10n$ minutes and work (including the writeup) until the timer goes off. Choose a textbook resource that you haven't used before and work through some problems or examples from that resource. State the problem(s) and write at least n sentences either solving the problem or writing what you tried.

▼ Choice Exercise 8 [7+6] (challenge)

1. [7] The group of $n \times n$ real-valued matrices with non-zero determinant under matrix multiplication is called

$$GL_n(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} \mid \det(A) \neq 0\}.$$

a. Let $GL_n^+(\mathbb{R})$ be the subset of $GL_n(\mathbb{R})$ which consists of matrices with *positive* determinant. Prove that $GL_n^+(\mathbb{R})$ forms a subgroup.

b. Prove that $GL_n^+(\mathbb{R})$ is a *normal* subgroup of $GL_n(\mathbb{R})$, compute the order of the quotient group $GL_n(\mathbb{R})/GL_n^+(\mathbb{R})$, and describe the coset that contains the matrix $-I_n$ (where I is the $n \times n$ identity matrix.)

c. Let $SL_n(\mathbb{R})$ be the subset of $GL_n(\mathbb{R})$ which consists of matrices with determinant one. Prove that $SL_n(\mathbb{R})$ is a subgroup of $GL_n(\mathbb{R})$.

d. Prove that $\{I, -I\} \leq SL_n(\mathbb{R})$, and moreover that $\{I, -I\} \lhd SL_n(\mathbb{R})$.

e. Give an example of two matrices that are in the identity coset of the quotient group $SL_n(\mathbb{R})/\{I, -I\}$, and give an example of two matrices that are in some other coset.

2. [6] We say that the *orthogonal group* $O(n, \mathbb{R})$ is the set of $n \times n$ matrices whose columns are orthonormal vectors, that is each column viewed as a vector has magnitude 1 and the dot product of any two distinct columns is zero.

a. Show that a matrix $A \in O(n, \mathbb{R})$ if and only if $A^T A = AA^T = I_n$.

b. Show that $O(n, \mathbb{R})$ is a group for all integers $n \geq 1$.

c. The *special orthogonal group* $SO(n, \mathbb{R})$ is the subset of $SO(n, \mathbb{R})$ consisting of all matrices whose determinant is positive. Prove that this is a group. (This is the group of rotations of the unit hypersphere in $\mathbb{R}^n$).

▼ Choice Exercise 9 [≤12] (proof)

In all parts, let $(G, *)$ be a group and H be a subgroup. You will be proving a fact about the binary operation on *cosets*

$$aH \bullet bH = (a * b)H.$$

1. [≤3] Start by giving an example of a case where this binary operation is *not* well-defined, by describing a group G a subgroup H, two cosets $aH = a'H$ and $bH = b'H$ such that $(a * b)H \neq (a' * b')H$.

(One point if you describe the example from class, two points if you describe a similar example, and three points if you describe a significantly different example.)

2. [3] Prove that if $\bullet$ is well-defined on the (left) cosets of H, then $gH = Hg$ for all $g \in G$.

3. [2] Prove that if $gH = Hg$ for all $g \in G$, then $\bullet$ is well-defined on the (left) cosets of H.

4. [1] Prove that if G is an abelian group, then every subgroup $H \leq G$ is a normal subgroup.

5. [3] Let $g \in G$ be some fixed element of G. Prove that a subgroup $H \leq G$ is a normal subgroup if and only if for all $g' \in gH$ there exists $h \in H$ such that $g' = h * g$.

▼ FINAL CHOICE EXERCISE [≤7]

Do any subset of the following parts, state which you did and how many points you are claiming for this problem (maximum of 7 points).

1. [≤3] Email Prof Kagey (pkagey@cpp.edu) or post in the `#feedback` channel on Discord with a typo that you have spotted in this assignment or with a suggestion about how to make something clearer. (One point for every six minutes that the typo got you stuck, with a maximum of three points.)

2. [≤2] Ask a question on Discord. (One point per question, up to 2.)

3. [≤4] Answer a classmate's question on Discord or give a hint about a problem on this week's assignment.[3] (Two points per answer/hint, up to 4.)

4. [≤4] Can you generalize any of the above problems in an interesting way? If so say what the generalization is, attempt to solve it, say something about it, and say how many points you feel like your effort was worth, up to 4.

5. [≤4] Choose one or more problems from our textbook or a previous homework assignment which interest you and write the problem statement along with an answer. Say how many points you feel like your effort was worth, up to 4.

6. [2] Stop by student drop-in hours.

7. [2 or 5] Work on at least one of the above exercises with a classmate. State which classmate you worked with and which exercises you collaborated on. If you've collaborated with someone for the first time, you get 5 points, if you've collaborated with someone before, you get 2 points for working with them again.

8. [2] Turn in the assignment on time (without requesting an extension).

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[1] This way of drawing a polyhedron is sometimes called a "Schlegel diagram." ↩

[2] When $n = 2$ this quotient group is called the projective linear group. ↩

[3] The syntax for a hint on Discord is double bars `||` on each side of the spoiler like this: `Remember that ||1 + x < x + 2||.` ↩

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